

Chemistry, 2016 EC, natural science, entrance exam, round 1, Tigray

Grade 12 · Chemistry · 2016 E.C.

Time allowed: 120 minutes**Questions:** 71**Total marks:** 71

Name _____ Grade _____ ID No. _____ Date _____

1. The number of significant figures obtained by addition of 300.53 and 2.507 is: (1)
A. 7 B. 5
C. 6 D. 4
2. Digits that make measurements certain plus one of uncertain values are known as _____. (1)
A. scientific notation B. decimal places
C. significant figures D. conversion factors
3. Which one of the following is the CORRECT description of uncertainty in measurements? It is used that exact number is (1)
A. whose values are known exactly B. accurate number
C. certain number D. whose values known inexactly
4. The spontaneous emission of high-energy radiation and particles is said to be: (1)
A. Radioisotope B. Radioactive decay
C. Radioactivity D. Radioactive radiation
5. "Both the position and the momentum of an electron in atom cannot be determined at the same time", this is the statement of the: (1)
A. Bohr's theory B. de-Broglie's statement
C. Planck's theory D. Heisenberg's uncertainty principle
6. Which of the following is CORRECT according to Hund's rule? (1)
A. The lowest energy electron configuration for degenerate orbitals is the one with the maximum number of unpaired valence electrons
B. No two electrons in a multi-electron atom have the same set of four quantum numbers; principal, secondary, magnetic and spin
C. The most stable atomic structures are those in which the electrons are in the lowest energy orbitals available
D. Electrons in an atom move in a circular orbits called stationary states and have a dual nature, particle and wave
7. Silver (Ag) has two stable isotopes; silver-107 (106.905 amu) that contributes 51.85% and silver-109 (108.905) contributes 48.15%. What is the relative atomic mass of silver? (1)
A. 106.868 B. 107.868
C. 108.868 D. 109.868
8. Which of the following is the class of an element with atomic number of 38? The element is: (1)
A. a transition element B. a representative element
C. an inner transition element D. post-transitional element

9. The average mass of the atoms in an element is known to be (1)

A. isotopic mass
B. molar mass
C. equivalent mass
D. atomic mass

10. Which of the following statements is CORRECT about electromagnetic radiation? (1)

A. The energy of an electromagnetic radiation increases with increase in its frequency
B. The energy of an electromagnetic radiation increases with increase in its wavelength
C. The speed of an electromagnetic radiation is inversely proportional to its wavelength
D. The wavelength of an electromagnetic radiation is directly proportional to its frequency

11. Which of the following is the MOST STABLE ground state electronic configuration of Molybdenum? ($Z = 42$) (1)

A. $[\text{Kr}] 5s^2 4d^4$
B. $[\text{Kr}] 5s^1 4d^5$
C. $[\text{Kr}] 5s^2 4d^3$
D. $[\text{Kr}] 5s^1 4d^6$

12. Which of the following is NOT responsible for the shape of the modern periodic table? (1)

A. valence electron
B. atomic mass
C. atomic number
D. the energy level

13. Which of the following is NOT included in the intermolecular forces found in covalent compounds? (1)

A. Dipole-Dipole Forces
B. London Forces
C. Dispersion Forces
D. Electrostatic Forces

14. Among the following properties given below which one is CORRECT about crystalline solid? (1)

A. Ionic crystals have weak inter-ionic forces, which makes them to have low melting points.
B. Molecular crystals are conductors of heat and electricity in molten state and usually have high melting points.
C. Covalent network crystals are conductors of heat and electricity and have low melting points.
D. Metallic crystals are ductile, malleable and they usually have high melting points.

15. Which of the following is observed in liquid water or solid ice due to hydrogen bonding? (1)

A. Molecules of water are closer together in ice than in liquid water.
B. At the melting point of water, ice is denser than liquid water.
C. Molecules of water are closer together in liquid water than in ice.
D. When ice and water are mixed, the water floats on the surface of the ice.

16. Among the following statements which explains the exception to the octet rule? When compound is formed, the central atom has: (1)

A. eight electrons configuration.
B. more than eight electron configuration.
C. core element electron configuration.
D. noble gas electron configuration.

17. The properties of a covalent molecule depend on the intermolecular forces within the molecule. Which of the following is an example that can represent these phenomena? (1)

A. Polar cis -1, 2-dichloroethene has a higher boiling point than non-polar trans -1, 2-dichloroethene.
B. Polar cis -1, 2-dichloroethene has a lower boiling point than non-polar trans -1, 2-dichloroethene.
C. Non-polar cis -1, 2-dichloroethene has a higher boiling point than polar trans -1, 2-dichloroethene.
D. Non-polar cis -1, 2-dichloroethene has a lower boiling point than polar trans -1, 2-dichloroethene.

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18. Describe the hybridization scheme for the central atom of tetrachloroiodate, radical, ICl_4^- . (1)
- A. dsp^2 B. sp^3d
C. sp^3d^2 D. d^3sp^2
-
19. "Reactant molecules are first transformed into an intermediate activated complex that is unstable and breaks into the products", this is called (1)
- A. reaction theory B. collision theory
C. activation theory D. transition state theory
-
20. Which of the following differentiates between biological catalysts and non-biological catalysts? Biological catalysts (1)
- A. are largely organic in nature while non-biological catalysts can be inorganic compounds.
B. are largely inorganic in nature while non-biological catalysts can be organic compounds.
C. speed up chemical reactions while non-biological catalysts inhibit chemical reactions.
D. inhibit chemical reactions while non-biological catalysts speed up chemical reactions.
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21. What is the difference between homogenous and heterogeneous reactions? A reaction between substances in the same (1)
- A. state is called a homogenous reaction while between different states is heterogeneous reaction.
B. state is called heterogeneous reaction while between different states is homogenous reaction.
C. phase is called a homogenous reaction while between different phases is heterogeneous reaction.
D. phase is called heterogeneous reaction while between different phases is homogenous reaction.
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22. Which of the following explains how transition state theory affects a reaction rate? (1)
- A. For reaction to occur, reactant molecules must collide with a certain minimum energy.
B. For reaction to take place, reactants must collide with the proper orientation.
C. The rate of reaction is directly proportional to the number of collisions per second.
D. The rate of the reaction is dependent on the speed at which the reactants can climb to the top of energy barrier.
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23. Rate constants of reactions are used when rates are determined experimentally. This is done by gathering data on the variation of (1)
- A. concentration over time. B. time over concentration.
C. speed with time. D. time with speed.
-
24. The decomposition of hydrogen iodide on gold surface is zero - order in HI. The rate constant for the reaction is 0.050 M/s . If the initial concentration of HI is 0.50 M , what is the half - life of this reaction? (1)
- A. 0.1 s B. 1.0 s
C. 0.5 s D. 5.0 s
-
25. An equation that defines the experimentally determined relation between the concentrations of reactants and rate of that reaction is known as _____. (1)
- A. rate law B. rate constant
C. order of reaction D. mechanism reaction
-
26. The ratio of the rate constant for the forward reaction over the reverse reaction is said to be _____. (1)
- A. gas constant B. rate constant
C. reaction constant D. equilibrium constant

27. Which of the following is CORRECT for a reaction at a chemical equilibrium? Given: Q_c is the reaction quotient and K_{eq} is the equilibrium constant (1)
- A. $Q_c = K_{eq}$ B. $Q_c > K_{eq}$
C. $Q_c \leq K_{eq}$ D. $Q_c \geq K_{eq}$
-
28. What is the value of equilibrium constant (K_p) for the following reaction? $2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$ Given: the equilibrium partial pressures of the three gases (SO_3 , O_2 and SO_2) are 0.10 atm, 0.25 atm and 0.50 atm, respectively. (1)
- A. 0.625 atm B. 6.25 atm
C. 62.5 atm D. 625 atm
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29. The equilibrium constant K_c for the equilibrium reaction $CO(g) + Cl_2(g) \rightleftharpoons COCl_2(g)$ is 6.1. In which direction will the reaction proceed to reach equilibrium? Given: $[CO] = 2.5\text{ M}$, $[Cl_2] = 1.0\text{ M}$ and $[COCl_2] = 5.0\text{ M}$ (1)
- A. Forward reaction B. Backward reaction
C. In the direction of the reactants D. Not possible to predict
-
30. Which of the following is the CORRECT order of carboxylic acid listed below based on the number of functional group that contains mono, di and tricarboxylic acids, respectively? (1)
- A. citric acid, acetic acid and oxalic acid B. oxalic acid, acetic acid and citric acid
C. acetic acid, citric acid and oxalic acid D. acetic acid, oxalic acid and citric acid
-
31. A triglyceride containing primarily unsaturated fatty acids is known as (1)
- A. oil B. ester
C. soap D. fat
-
32. Which of the following methods is used for the preparation of saturated monocarboxylic acids? Oxidation of (1)
- A. primary alcohols B. secondary alcohols
C. tertiary alcohols D. quaternary alcohols
-
33. Which of the following is the common use of esters? It is used for (1)
- A. making soaps, detergents and shampoos B. making artificial flavors and perfumes
C. preparation of carboxylic acids D. preparation of primary alcohols
-
34. A student placed 50 mL of 1 M sulfuric acid into the 500 mL round - bottom flask. He/she added 5 g of sodium dichromate and a few pieces of broken porcelain. After mixing the resulting solution, he/she cooled the mixture under a running tap. He/she added 5 mL of ethanol drop-wise into the flask and boiled under reflux for 20 minutes. In this experiment ethanol (1)
- A. is oxidized by sodium dichromate and produces acetic acid
B. is reduced by sodium dichromate and produces acetic acid
C. reacts with sodium dichromate to produce ethane, carbon dioxide and water
D. reacts with sodium dichromate to produce an ester and water
-
35. "At a given temperature, the solubility of a sparingly soluble and chemical unreactive gas in a liquid is proportional to the partial pressure of the gas over the solution", this quantitative relationship is given by _____. (1)
- A. Raoult's law B. Hess's law
C. Henry's law D. Dalton's law

-
- 36.** The ratio of the number of moles of a component in a mixture to the total number of moles in the mixture is known to be _____. (1)
- A. molarity
B. molality
C. mole percent
D. mole fraction
-
- 37.** Which of the following distinguishes between homogenous and heterogeneous mixtures? In homogenous mixture (1)
- A. components are not distributed uniformly while in heterogeneous mixture the components are distributed uniformly
B. it contains distinct regions of different composition while in heterogeneous mixture the components do not have visible boundary
C. components are distributed uniformly while in heterogeneous mixture the components are not distributed uniformly
D. composition is not uniform while in heterogeneous mixture the components are mixed as individual atoms and molecules
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- 38.** Which of the following explanations differentiates between suspensions and colloids? (1)
- A. Suspensions are composed of particles that settle down in due course of time while colloidal particles do not settle down
B. Suspensions are homogenous mixtures composed of particles visible to the naked eye while colloids are heterogeneous mixtures.
C. Colloids are composed of particles that settle down in due course of time while suspensions do not settle down.
D. Colloids are homogenous mixtures composed of particles visible to the naked eye while suspensions are heterogeneous mixtures.
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- 39.** How much water is added to 75 mL of 0.5 M HCl solution to prepare a 0.1 M HCl solution? (1)
- A. 75 mL
B. 150 mL
C. 300 mL
D. 375 mL
-
- 40.** Which of the following methods is used to prepare a supersaturated sodium thiosulfate solution? (1)
- A. Slowly dropping the temperature of a saturated solution of sodium thiosulfate
B. Increasing the volume of unsaturated solution through slow evaporation
C. Slowly increasing the temperature of unsaturated solution of sodium thiosulfate
D. Increasing the volume of a saturated solution through slow evaporation
-
- 41.** The maximum amount of a solute that dissolves in a given quantity of solvent at a given temperature is said to be _____. (1)
- A. molarity
B. molality
C. normality
D. solubility
-
- 42.** The ratio of the number of equivalents of the solute divided by the total volume of solution gives _____. (1)
- A. equivalent mass
B. molar mass
C. normality
D. molality

43. Which of the following is CORRECT about the heat of solution? (1)
- A. If the solute-solvent attraction is stronger than both the solvent-solvent and solute-solute attraction, the solution process is endothermic.
 - B. If the solute-solvent attraction is stronger than both the solvent and solute-solute attraction, the solution process is exothermic.
 - C. If the solute-solvent attraction is stronger than both the solvent-solvent and solute-solute attraction, the solution process is neutral.
 - D. The solution process is independent of solute-solvent, solvent-solvent or solute-solute attractions.
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44. Which of the following is the definition of an acid according to Lewis concept? An acid is a substance that: (1)
- A. increases $[H^+]$ when placed in water
 - B. decreases $[OH^-]$ when placed in water
 - C. acts as an electron pair acceptor
 - D. acts as a proton donor
-
45. Which one of the following is a buffer solution? A mixture of (1)
- A. ethanoic acid and ethanol
 - B. acetic acid and sodium acetate
 - C. acetic anhydride and acetate ion
 - D. sodium ethanoate and ethyl alcohol
-
46. Which of the following determines the equivalent mass of acids and bases? (1)
- A. Total number of electrons transferred in a chemical reaction
 - B. Number of hydrogen ions or hydroxide ions transferred in a chemical reaction
 - C. Mole of reactants and products at the equilibrium point of the reactions
 - D. Mole of reactants and products at the start and at the end of a chemical reaction
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47. What is the concentration of OH^- ions in a solution whose $pH = 6$? (1)
- A. $1.0 \times 10^{-6} M$
 - B. $1.0 \times 10^{-8} M$
 - C. $1.0 \times 10^{-6} M$
 - D. $1.0 \times 10^{-8} M$
-
48. The pH of a $1.00 M$ solution of formic acid ($HCOOH$) is 1.88 . What is the percent ionization of formic acid? K_a of $HCOOH = 1.7 \times 10^{-4}$ (1)
- A. 0.013
 - B. 1.30
 - C. 0.13
 - D. 13.0
-
49. What is the $[H^+]$ ion concentration in $1.0 \times 10^{-3} M$ solution of a weak acid (CH_3COOH), whose $K_a = 1.8 \times 10^{-5}$ (1)
- A. $4.13 \times 10^{-4} M$
 - B. $1.34 \times 10^{-4} M$
 - C. $4.13 \times 10^{-5} M$
 - D. $1.34 \times 10^{-5} M$
-
50. From the following alternatives, which one is a Brønsted-Lowry base? (1)
- A. NH_3
 - B. NH_4^+
 - C. H_2S
 - D. H_3O^+
-
51. Which of the following is a CORRECT explanation for the existence of hydrogen ion (H^+) as hydronium ion (H_3O^+) in water? (1)
- A. The small size and high charge of protons
 - B. The large size and small charge of protons
 - C. The neutral pH of liquid water
 - D. The small reactivity of water with protons

52. The difference between the reduction potentials of a voltaic cell's cathode and anode is known as _____. (1)

- A. electrode cell
B. electrochemical cell
C. electrochemical potential
D. cell potential

53. Which of the following is galvanic requires continuous supply of reactant and continuous removal of product to keep functioning? (1)

- A. Daniel cell** **B. Leclanché Cell**
C. Fuel Cell **D. Mercury Cell**

54. Which of the following factors DOES NOT affect the preferential discharge of ions in the electrochemical series? (1)

- A. The nature of the electrodes
- B. The nature of the ions in the electrolyte
- C. The concentration of the ions in the electrolyte
- D. The position of the ions in the electrochemical series

55. During the electrolysis of dilute sulfuric acid solution, which of the following is discharged at the anode electrode? (1)

- A. OH⁻ ions are discharged in preference to SO₄²⁻ ions, because the concentration of OH⁻ ions is greater than SO₄²⁻ ions.
- B. SO₄²⁻ ions are discharged in preference to OH⁻ ions, because the concentration of SO₄²⁻ ions is greater than OH⁻ ions.
- C. OH⁻ ions are discharged in preference to SO₄²⁻ ions, because the OH⁻ ion is below the SO₄²⁻ ion in the electrochemical series.
- D. SO₄²⁻ ions are discharged in preference to OH⁻ ions, because SO₄²⁻ ions are stronger bases than OH⁻ ions.

56. An electrolysis reaction of copper (II) sulfate deposited copper metal at the cathode and liberated oxygen gas at the anode. What are the electrodes used for this reaction?

- A. Zinc electrodes
B. Magnesium electrodes
C. Copper electrodes
D. Platinum electrodes

57. Which of the following is CORRECT about the environmental effects of corrosion? (1)

- A. is a non-redox reaction of metals
- B. is the major factor to prevent rusting of iron
- C. causes metals to be stronger in industry and construction
- D. causes enormous damage to buildings, bridges, ships and cars

58. Which of the following is the CORRECT half reactions at the electrodes and overall reaction observed during the electrolysis of CuSO_4 solution using inert electrodes? (1)

- A.** Anode: $2\text{Cu}^{2+}(\text{aq}) + 4\text{e}^{-} \rightarrow 2\text{Cu}(\text{s})$ Cathode: $4\text{OH}^{-}(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g}) + 4\text{e}^{-}$ Cell reaction: $2\text{Cu}^{2+}(\text{aq}) + 4\text{OH}^{-}(\text{aq}) \rightarrow 2\text{Cu}(\text{s}) + 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$
- B.** Anode: $4\text{OH}^{-}(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g}) + 4\text{e}^{-}$ Cathode: $2\text{Cu}^{2+}(\text{aq}) + 4\text{e}^{-} \rightarrow 2\text{Cu}(\text{s})$ Cell reaction: $2\text{Cu}^{2+}(\text{aq}) + 4\text{OH}^{-}(\text{aq}) \rightarrow 2\text{Cu}(\text{s}) + 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$
- C.** Anode: $\text{Cu}(\text{s}) \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{e}^{-}$ Cathode: $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{Cu}(\text{s})$ Cell reaction: $\text{Cu}(\text{s}) + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Cu}^{2+}(\text{aq}) + \text{Cu}(\text{s})$
- D.** Anode: $2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g}) + 4\text{e}^{-} \rightarrow 4\text{OH}^{-}(\text{aq})$ Cathode: $2\text{Cu}(\text{s}) \rightarrow 2\text{Cu}^{2+}(\text{aq}) + 4\text{e}^{-}$ Cell reaction: $2\text{Cu}(\text{s}) + 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g}) \rightarrow 2\text{Cu}^{2+}(\text{aq}) + 4\text{OH}^{-}(\text{aq})$

59. What mass of aluminum is deposited electrolytically in 10 minutes by a current of 40 A? ($F = 96500 \text{ C/mol}$) (1)

- A. 2.25 g B. 0.24 g
C. 6.71 g D. 0.67 g

60. Which of the following is the CORRECT sequence of the first five most abundant elements in the earth's crust? (1)

- A.** oxygen, aluminum, silicon, iron and calcium
B. oxygen, silicon, iron, aluminum and calcium
C. oxygen, iron, silicon, calcium and aluminum
D. oxygen, silicon, aluminum, iron and calcium

61. Which compound of chromium is used in the production of stainless steel? (1)

- A. Nichrome
B. Chromate ion
C. Ferrochrome
D. Dichromate ion

62. When copper powder is used as a catalyst, silicon reacts with methyl chloride (CH_3Cl) at high temperature. (1)
What is the product of this reaction?

- A. CH₃Cl B. CH₃SiCl
C. (CH₃)₂SiCl₂ D. (CH₃)₃SiCl₂

63. Which of the following is the procedure for the manufacture of tin? (1)

- A.** Leaching the mineral cassiterite with acid to remove impurities followed by roasting of with carbon in furnace at high temperature.
- B.** Electrochemical reduction of the mineral cassiterite after leaching the mineral ore.
- C.** Oxidation of the mineral cassiterite with hydrogen in furnace at low temperature followed by refining with acid to remove impurities.
- D.** Reduction of the mineral cassiterite with hydrogen in furnace at low temperature followed by refining with acid to remove impurities.

64. Which of the following IS NOT made by an addition polymer? (1)

- A. polyethylene
B. Teflon
C. Polystyrene
D. Polyester

65. Which of the following polymers is a natural polymer? (1)

- A. Glycogen
B. Nylons
C. Polystyrene
D. Polyester

66. "Entropy of an isolated thermodynamic system always increases during a spontaneous process", this is called (1)

- A.** Hess's law
B. The second law of thermodynamics
C. The first law of thermodynamics
D. The third law of thermodynamics

67. What are the monomers of Bakelite, in a condensation polymer? (1)

- A. Hexamethylene diamine and adipic acid
B. Terephthalic acid and ethylene glycol
C. Styrene and 1,3-butadiene
D. Formaldehyde and phenol

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68. Which of the following expression about entropy or entropy change is CORRECT? (1)
- A. Entropy is the sum of internal energy and the pressure-volume product of a system.
 - B. Entropy change for a chemical change is the product of the entropies of the products and the entropies of the reactants.
 - C. Entropy is the thermodynamic property of a system that is related to its degree of randomness.
 - D. Entropy change for a chemical change is the sum of the entropies of the reactants minus the sum of the entropies of the products.
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69. What is the standard free energy change, ΔG° , for the complete combustion of methane? $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$ Given: The standard free energies of formation: - $\text{CH}_4(\text{g}) = -51 \text{ kJ/mol}$ - $\text{CO}_2(\text{g}) = -394 \text{ kJ/mol}$ - $\text{H}_2\text{O}(\text{l}) = -237 \text{ kJ/mol}$ (1)
- A. -631 kJ/mol
 - B. -632 kJ/mol
 - C. -838 kJ/mol
 - D. -817 kJ/mol
-
70. A process that occurs only as long as energy is continually applied to the system is said to be (1)
- A. thermodynamic process
 - B. nonspontaneous process
 - C. spontaneous process
 - D. equilibrium process
-
71. What kind of system is formed by closing a flask containing water, so that NO water vapor can escape from or condense into the container? (1)
- A. An open system
 - B. A closed system
 - C. An isolated system
 - D. An adiabatic system

— END OF EXAMINATION —